Representational Geometry in Neural Networks

From Vision Transformers to RLHF-Aligned Language Models

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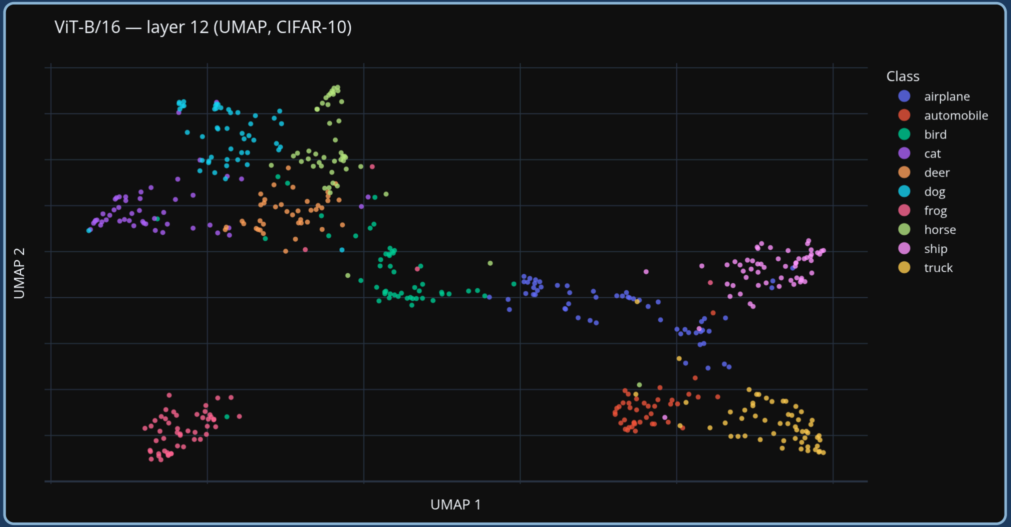


ViT-B/16 · layer 12 · CIFAR-10 · real data

Core Questions

Part 1 — Cross-modal alignment

When a vision model and a language model train on the **same task**, does a shared representational structure emerge across modalities?

- **Hypothesis A:** cross-modal signal is new information ⟶ improves model
- **Hypothesis B:** models converge regardless — extra signal adds little

CKA before/after cross-modal fine-tuning · DINOv2 + Llama · MS-COCO

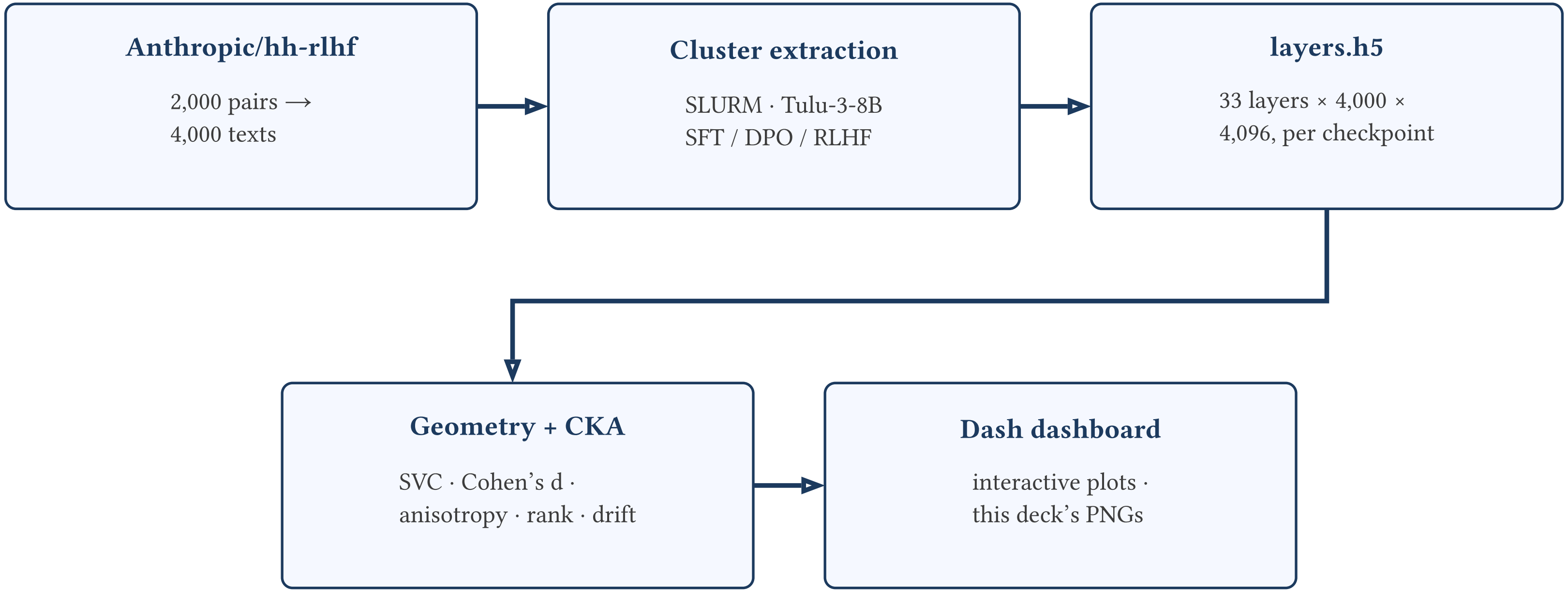
Part 2 — RLHF geometry

Does RLHF geometrically transform a language model's embedding space — encoding human preference as a **linearly separable structure**?

- Does alignment progressively separate chosen/rejected responses across SFT ⟶ DPO ⟶ RLHF?
- How does separation evolve **layer by layer**?

UMAP + LinearSVC per layer · Tulu-3-8B SFT/DPO/RLHF · `Anthropic/hh-rlhf`

Part 2 — Pipeline: From Dataset to Dashboard



`scripts/extract_embeddings.py` → `app/rlhf_geometry_compute.py` / `app/vision_compute.py` → `app/app.py`

Part 2 — What Each Checkpoint Actually Is

Tulu-3-8B alignment trajectory — 3 public AllenAI checkpoints, same base model, extracted at each post-training stage

SFT

`Llama-3.1-Tulu-3-8B-SFT`

- Instruction fine-tuning
- No preference signal yet

DPO

`Llama-3.1-Tulu-3-8B-DPO`

- Chosen > rejected, directly
- No reward model

RLHF

`Llama-3.1-Tulu-3-8B`

- Final RL stage (Tulu-3′s RLVR)
- Labeled **RLHF** for Part 1 consistency

Remark All 3 are the **same** underlying 8B model at 3 successive training stages — not 3 different architectures. This is what makes the SFT ⟶ DPO ⟶ RLHF comparison a clean trajectory rather than a cross-model comparison.

Definition: Linear CKA For column-centered X, Y:

$$\mathrm{CKA}(X,Y) = \frac{\left\|X^\top Y\right\|_F^2}{\left\|X^\top X\right\|_F \cdot \left\|Y^\top Y\right\|_F}$$

1 = proportional representations (no drift), 0 = orthogonal.

Definition: Cohen's d Direction $v = \frac{\mu_{\text{chosen}} - \mu_{\text{rejected}}}{\left\|\mu_{\text{chosen}} - \mu_{\text{rejected}}\right\|}$:

$$d = \frac{\left(\mu_{\text{chosen}} - \mu_{\text{rejected}}\right) \cdot v}{\sigma_{\text{pooled}}}$$

Effect size along the best linear separating direction.

Definition: Anisotropy Over randomly sampled pairs $i \neq j$:

$$\mathrm{aniso}(X) = \mathbb{E}_{i \neq j}\left[\cos\left(x_i, x_j\right)\right]$$

Near 1 = representations collapse to a narrow cone.

Definition: Effective rank Participation ratio of covariance eigenvalues λ_i:

$$\mathrm{erank}(X) = \frac{\left(\sum_i \lambda_i\right)^2}{\sum_i \lambda_i^2}$$

How many dimensions are effectively in use.

Tulu-3-8B SFT ⟶ DPO ⟶ RLHF · `Anthropic/hh-rlhf` · 4,000 samples · 33 layers

Negative finding

- Not linearly encoded, any stage
- LinearSVC peak: 0.543 / 0.544 / 0.550
- Near chance, all 3 checkpoints (layer 14)
- Cohen's d peak: 0.183 ⟶ 0.193 ⟶ 0.194
- No phase transition in preference geometry

But the geometry does shift

- Linear CKA, same inputs, layer by layer
- CKA(SFT, DPO) drops to **0.976** by layer 32
- Most of the shift happens SFT⟶DPO
- CKA(DPO, RLHF) stays near **0.999**
- RLHF/PPO barely moves geometry further

Full curves in the dashboard · Part 2 tab

Remark Peak accuracy near 0.55 matches published `hh-rlhf` baselines (0.57–0.66) and the dataset's own 63% label-agreement ceiling (Bai et al., 2022).

Reading the 0.55 baseline before the flat curves show up

What caps the accuracy?

- Bai et al. (2022): 63% researcher–crowdworker agreement
- Stiennon et al. (2020): 77%±2% with careful onboarding
- Label quality is a dataset property, not a constant
- Reward models on `hh-rlhf`: 57–66%, rarely above 72%

What that means for our 0.55

- Can't beat the label consistency of the data
- 0.55 is low, not far outside published baselines
- Real finding: the **missing layer trend**
- Not the raw accuracy number itself

Sources: Bai et al. 2022 · Stiennon et al. 2020 · reward-model benchmarks on `hh-rlhf`

Part 2 — "No Linear Separation": an RLHF Finding, or a DPO Finding?

Six objectives, side by side

- Sinha et al. (2026, `arXiv:2606.09850`): layer-wise probing, 6 objectives
- **KTO, GRPO** build linear separability (constructive)
- **DPO, ORPO** degrade it (geometric rotation)
- **PPO, SimPO** largely preserve baseline geometry

- Our RLHF checkpoint ≈ RLVR/PPO-like ⟶ fits "preserves" pattern
- Our DPO step: biggest CKA jump, **no** separability gain
- Fits "degrades via rotation," not "constructs a new axis"

> **Remark** Our negative finding is likely specific to DPO/PPO-style objectives — not to "RLHF" in general.

Source: Sinha et al. 2026 (`arXiv:2606.09850`) — very recent preprint, re-check before citation

Part 2 — Methodology Check: Pooling Decides a Lot

Approach	What's measured	Typical accuracy
Single-point probe (our approach)	Chosen/rejected as independent points, last-token pooled	0.5–0.6
Paired-difference probe	chosen − rejected per prompt pair, removes prompt variance	0.84–0.86 (`arXiv:2604.09870`)

- Our LinearSVC fights prompt-to-prompt variance
- Paired-difference removes that variance
- Not a weakness — a deliberately stricter question
- "Readable in raw space?" vs. "readable given pairs?"

> **Remark** Next step: re-run a paired-difference probe as a robustness check to see whether separability rises once prompt variance is removed.

Source: Kirin 2026 (`arXiv:2604.09870`) — very recent preprint, re-check before citation

Part 2 — CKA Drift Pattern: Not a One-Off

Independent cross-domain evidence for “SFT is the real turning point”

Language, layer-wise SFT

- `arXiv:2604.11838`
- Final layers far more sensitive to SFT
- Early/middle layers stay stable
- Matches our SFT⟶DPO jump

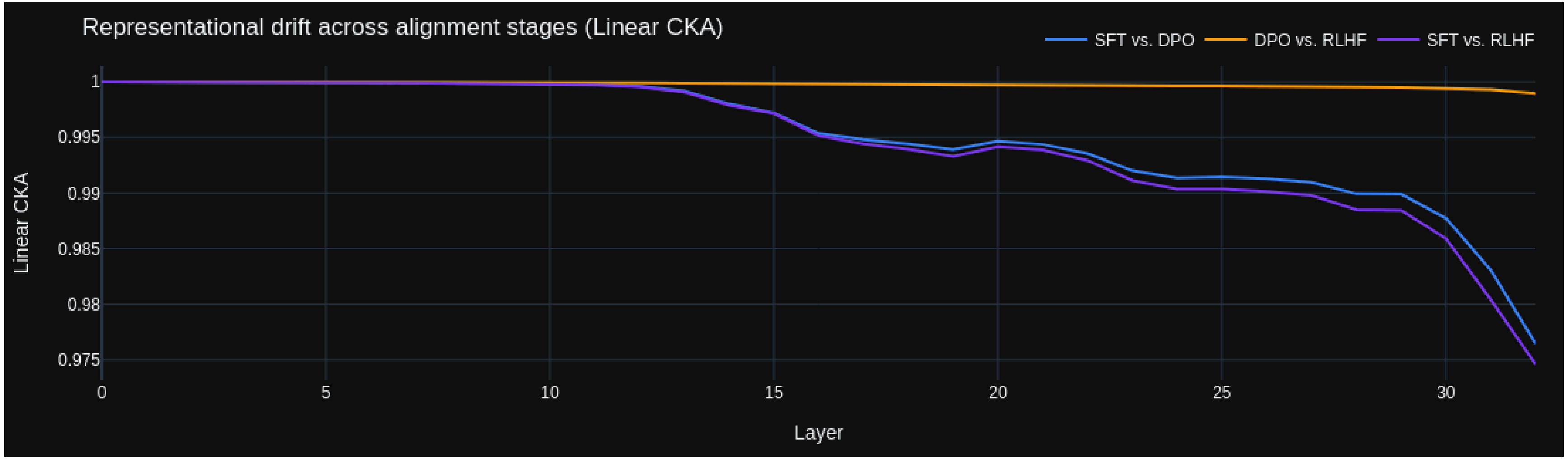
Speech domain, GRPO vs. SFT

- `arXiv:2607.08409`
- GRPO changes behavior, not representations
- Late RL step ≈ small geometry change
- Consistent with our DPO⟶RLHF pattern

Our “SFT⟶DPO is the turning point” finding rhymes elsewhere — strengthens the interpretation.

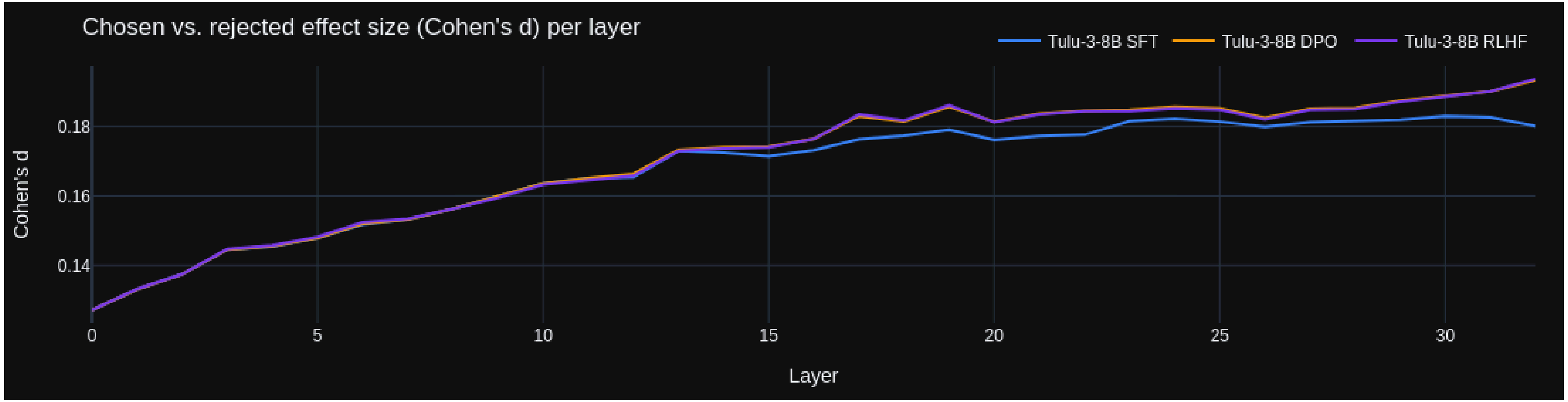
Sources: Zhao et al. 2026 (`arXiv:2604.11838`) · Kumar et al. 2026 (`arXiv:2607.08409`) — very recent preprints, re-check before citation

Part 2 — CKA Drift



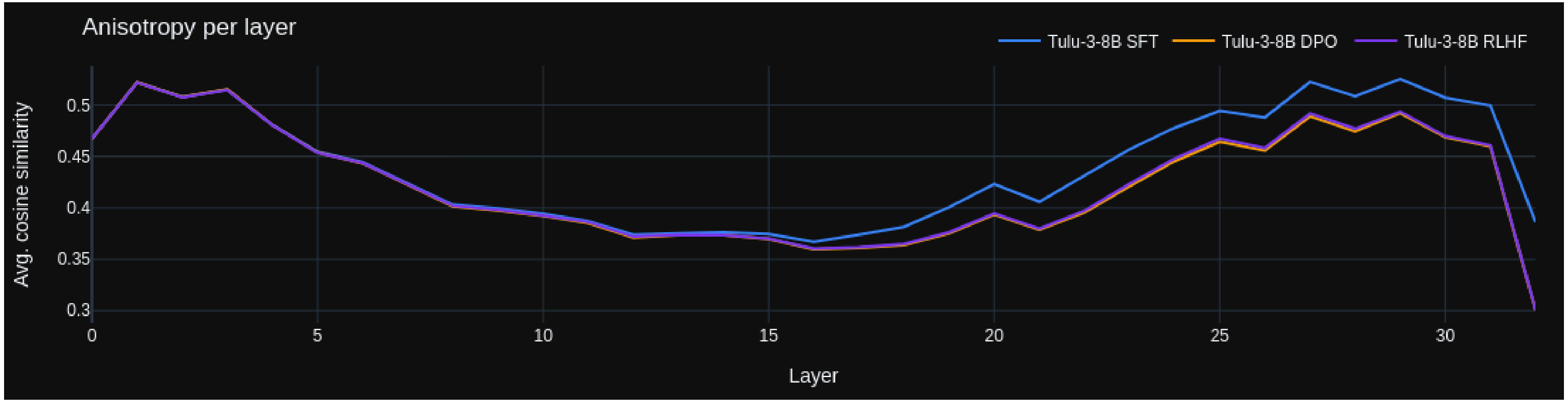
CKA drift — shift concentrated SFT to DPO, later layers

Part 2 — Cohen's d



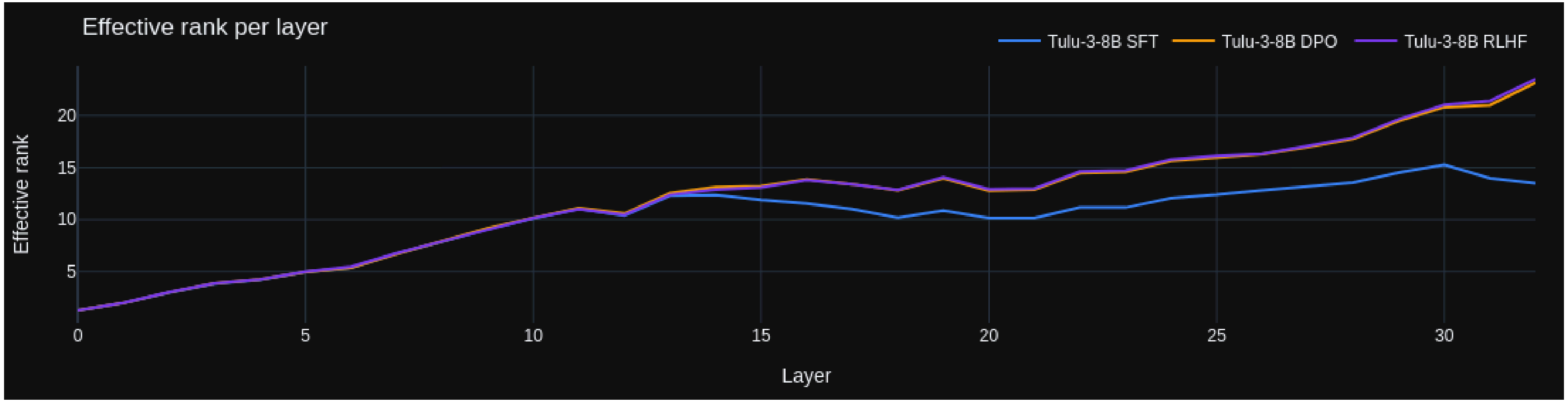
Cohen's d — small, flat across all layers

Part 2 — Anisotropy



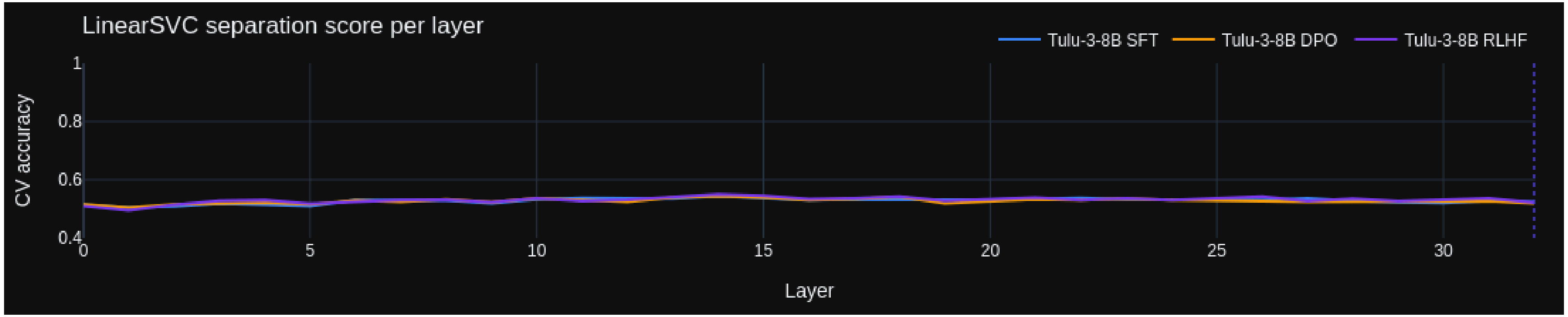
Anisotropy — dips mid-stack, rises again late

Part 2 — Effective Rank



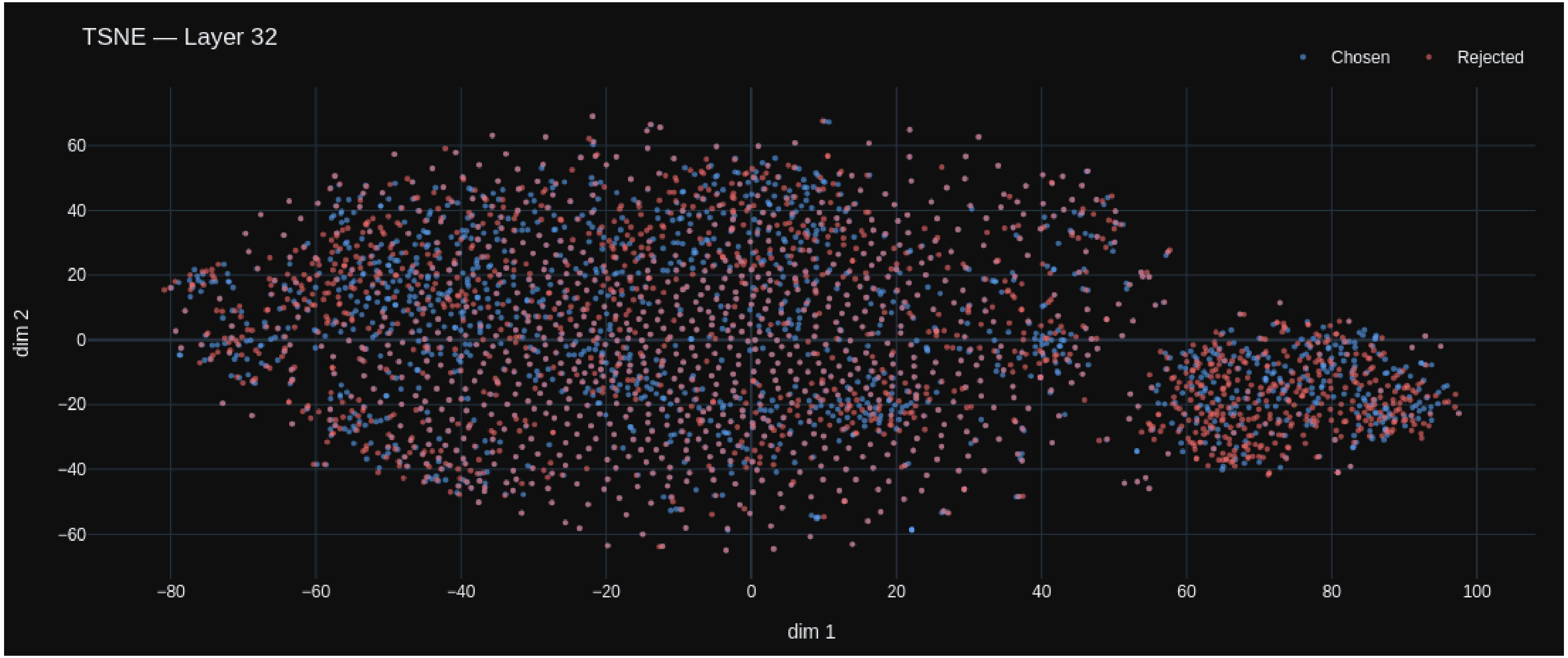
Effective rank — dimensionality usage per layer

Part 2 — LinearSVC Accuracy



LinearSVC accuracy — pinned near chance (0.5)

Part 2 — t-SNE Layer 32



t-SNE, layer 32 — chosen/rejected heavily overlap

Insight: no readable linear "preference direction"

- Flat, near-chance at **every** layer — all 3 checkpoints
- Peak SVC: 0.543 / 0.544 / 0.550
- DPO/PPO preserve or degrade separability; KTO/GRPO construct it
- Looks objective-specific, not universal to "RLHF"

Insight: but geometry is not static

- Real structural reorganization, concentrated at **SFT → DPO**
- CKA(SFT, DPO) drops to 0.976; CKA(DPO, RLHF) stays near 0.999
- Most "alignment work" happens before PPO

> **Remark** Anisotropy and effective rank shift in that same window — one transition, not three.

*Alignment reshapes geometry **globally**, not a single axis. Lee et al. (2024): DPO **bypasses** a capability rather than erasing it.*

Part 1 — Cross-modal Alignment · Outlook

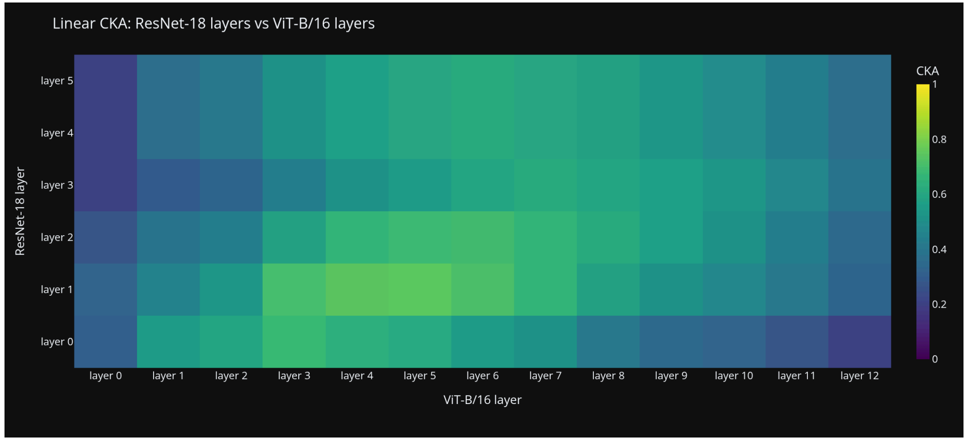
Setup · DINOv2 + Llama · MS-COCO · multi-class prediction

Both models trained on **same task** (object category prediction) · measure CKA · fine-tune with loss term encouraging similarity to the other model's representation · compare

CKA high before fine-tuning: models converge inherently **(Platonic Representation Hypothesis)**

CKA rises after fine-tuning: cross-modal signal is genuinely new information

Layer-wise CKA map shows **where** shared meaning lives · UMAP + dashboard to visualize

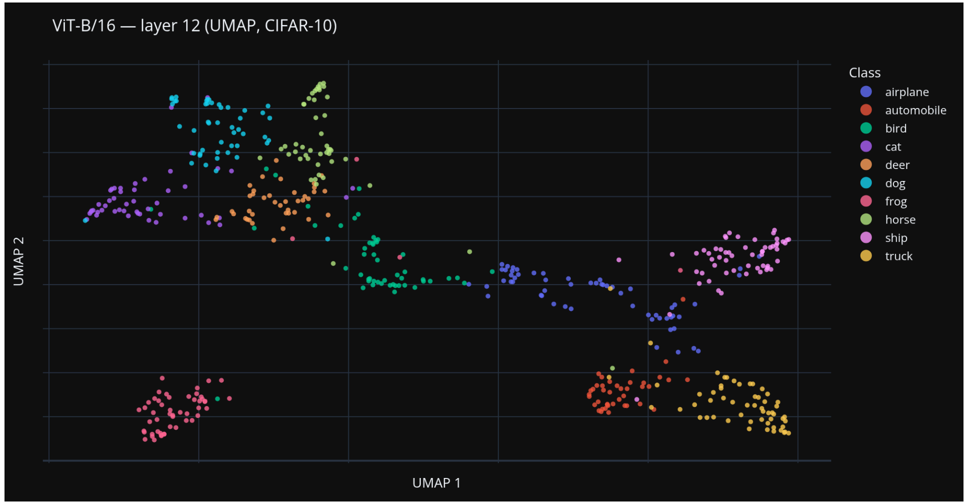


Interactive Plotly Dash · CKA heatmap

Next steps — toward final

- Cross-modal CKA: DINOv2 vs. Llama on MS-COCO pairs
- **CLIP (ViT-B/32)** as upper bound — explicitly trained cross-modal alignment
- Compare base vs. Instruct CKA scores

Deliverable & Next Steps



Interactive Plotly Dash · CKA heatmap · UMAP layer slider · LinearSVC score

Project deliverable

The dashboard **is** the deliverable — interactive visualization of all findings: layer slider, CKA heatmaps, UMAP scatter, SVC score

Perspective

Extend to RLHF training **snapshots** — plot cross-modal CKA vs. training step to see if alignment shifts **during** RLHF

References

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arXiv:2607.08409 · **GRPO changes behavior, not geometry · re-check**